

Feature Scaling: Normalization

Module 25 Study Notes • Feature Engineering

1. What is Normalization?

Normalization is the process of scaling numerical input features to a fixed range, typically **[0, 1]**, without distorting the differences in the ranges of values. This is crucial for eliminating the impact of units (e.g., converting weight from grams or pounds to a scale of 0-1) so that algorithms can process the data more effectively.

2. Primary Normalization Techniques

A. Min-Max Scaling

This is the most widely used normalization method. It maps every value in a column to a range between 0 and 1.

Formula: $x_{scaled} = (x - x_{min}) / (x_{max} - x_{min})$

If x is the minimum, the result is 0; if it is the maximum, the result is 1.

B. Mean Normalization

Centers the data around 0 by subtracting the mean and dividing by the range.

Formula: $x_{scaled} = (x - mean) / (x_{max} - x_{min})$

C. Max-Abs Scaling

Scales each feature by its maximum absolute value. This is particularly useful for **Sparse Data** (datasets with many zeros) because it preserves the sparsity (the zeros remain zeros).

Formula: $x_{scaled} = x / |max(x)|$

D. Robust Scaling

This method uses statistics that are robust to outliers, specifically the **Median** and the **Interquartile Range (IQR)**.

Formula: $x_{scaled} = (x - median) / (Q3 - Q1)$

Since the median and IQR are not significantly affected by extreme outliers, this technique is preferred when your data is "noisy."

3. Practical Tips for Choosing

- **Standardization (Z-score)** is the default choice for most algorithms, especially those involving distances (KNN) or gradients (Neural Networks).
- **Min-Max Scaling** is best when you have a predefined range (e.g., image pixel values are always 0-255).
- **Robust Scaling** is the go-to when your data contains many outliers that you cannot or choose not to remove.
- **When in doubt, experiment!** Scaling is not a "one-size-fits-all" solution. Run your model with different scaling techniques and choose the one that provides the best validation score.